Early-life stress: epigenetic variations and depression

Abstract

Early-life stress (ELS)—including physical, emotional, or sexual abuse and neglect—severely impacts cognitive and behavioral development. ELS is linked to epigenetic alterations, such as DNA methylation and altered microRNA (miRNA) expression, which increase the risk of developing psychiatric disorders in adulthood. Notably, individuals exposed to multiple ELS events face a fourfold higher rBachelorEnglishBachelor'sBachelor'sisk of developing depression.

This review focuses on the epigenetic dysregulation of oxytocin and serotonin signaling pathways following ELS. Clinical studies in adolescents show that ELS alters specific circulating miRNAs (e.g., miR-450a-2-3p, miR-556-3p) and increases oxytocin gene methylation, which correlates with reduced neural responses (N170) to social stimuli. Similarly, animal models of maternal separation demonstrate altered expression of serotonin-related genes (*Htr1a*, *Slc6a4*) and target miRNAs in the prefrontal cortex, while highlighting that environmental enrichment can potentially reverse these epigenetic changes.

Overall, understanding these epigenetic mechanisms is crucial for identifying reliable biomarkers and developing targeted therapeutic and preventive strategies for depression.